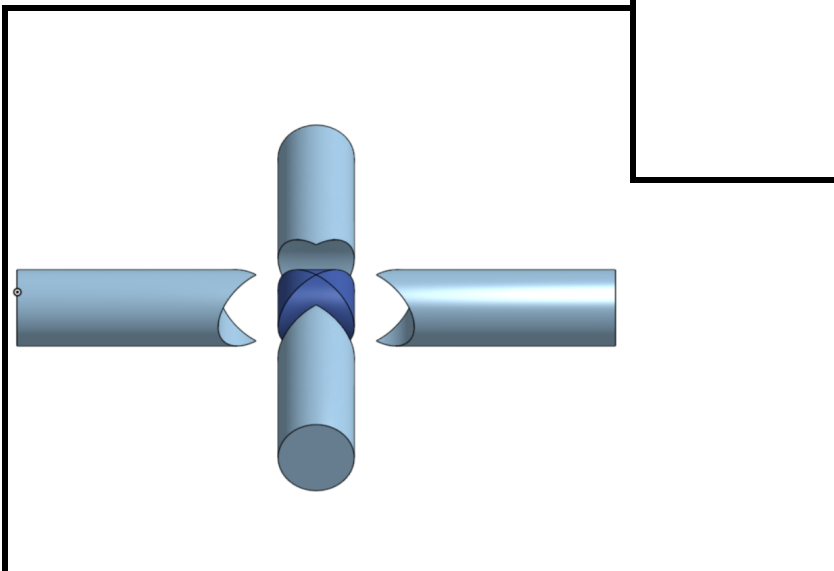
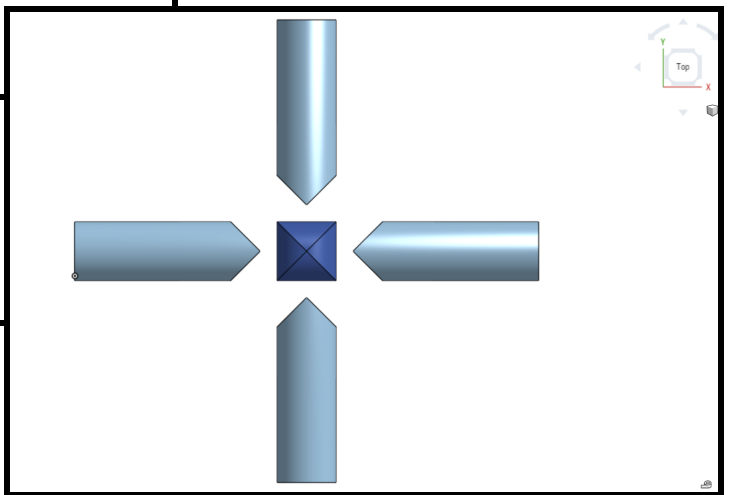
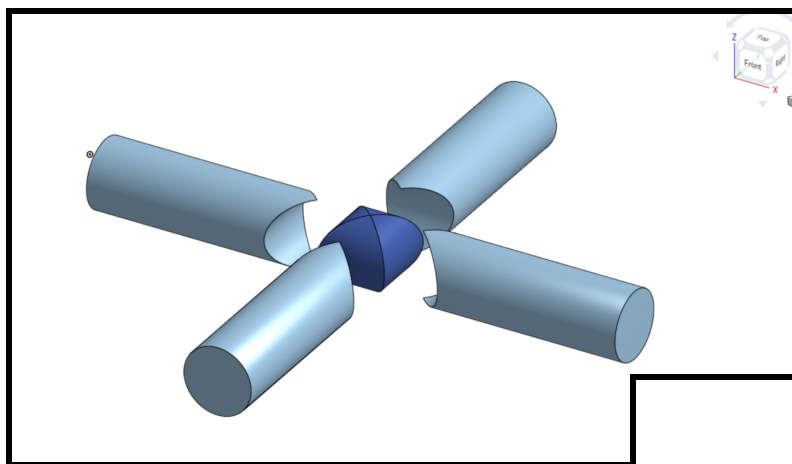


2D/3D Pipe Intersection

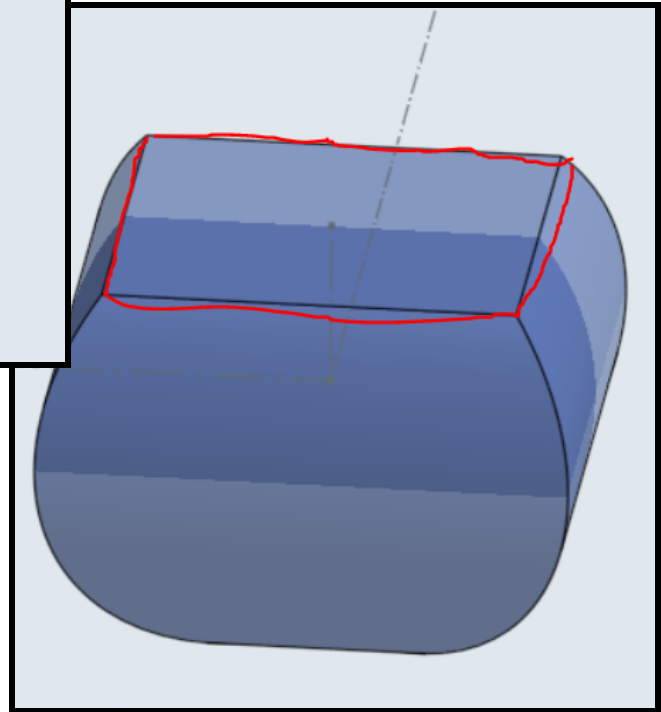
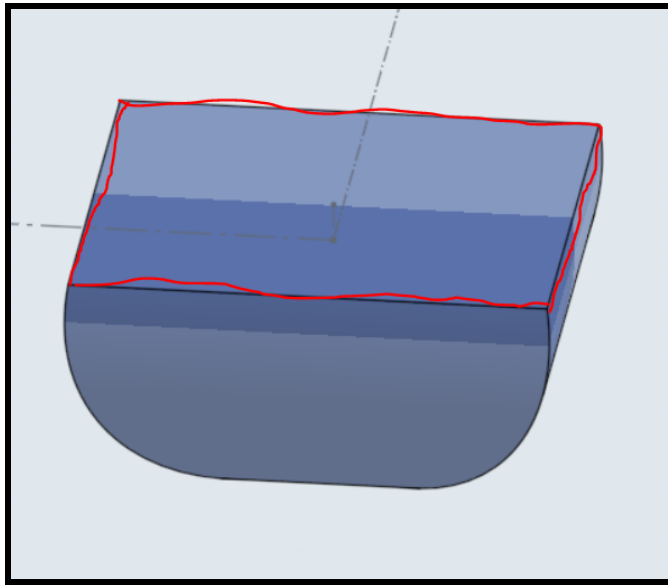
Pipe Diameter = 1.27cm
(Radius = 0.635cm)

Pipe Length = 7.976cm

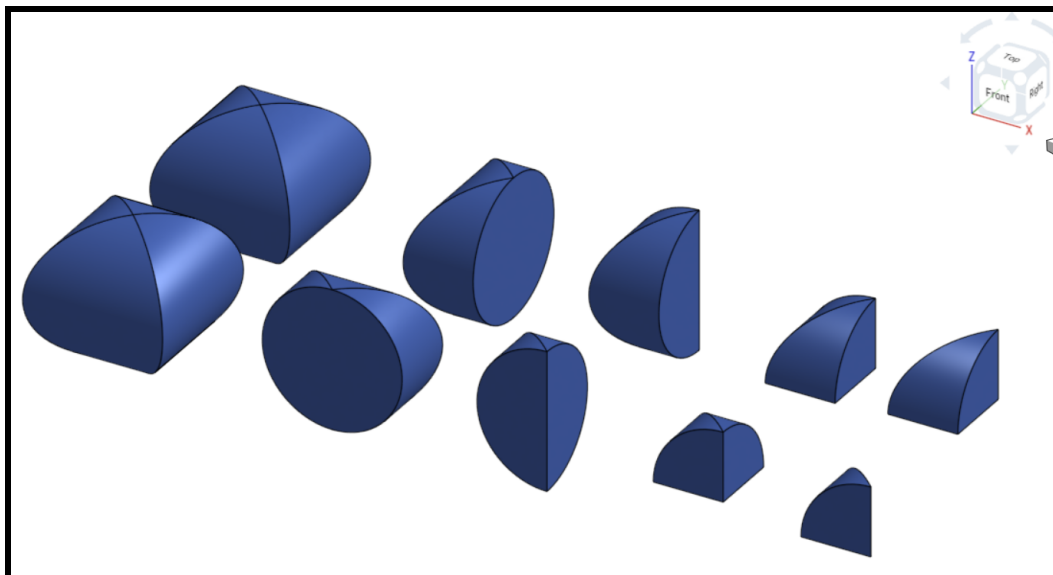
2D Intersect Slideaway Images...



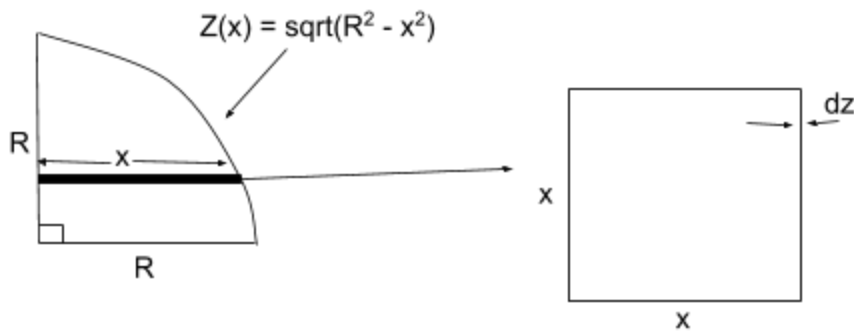
2D Intersect Square Cross-section Images (Shows that the Cross Section of the Intersect Volume Yields Squares)...



2D Intersect Slice Up...



2D Intersect Analytical Volume Calculation...



$$z(x) = \sqrt{R^2 - x^2}$$

$$z^2 = R^2 - x^2$$

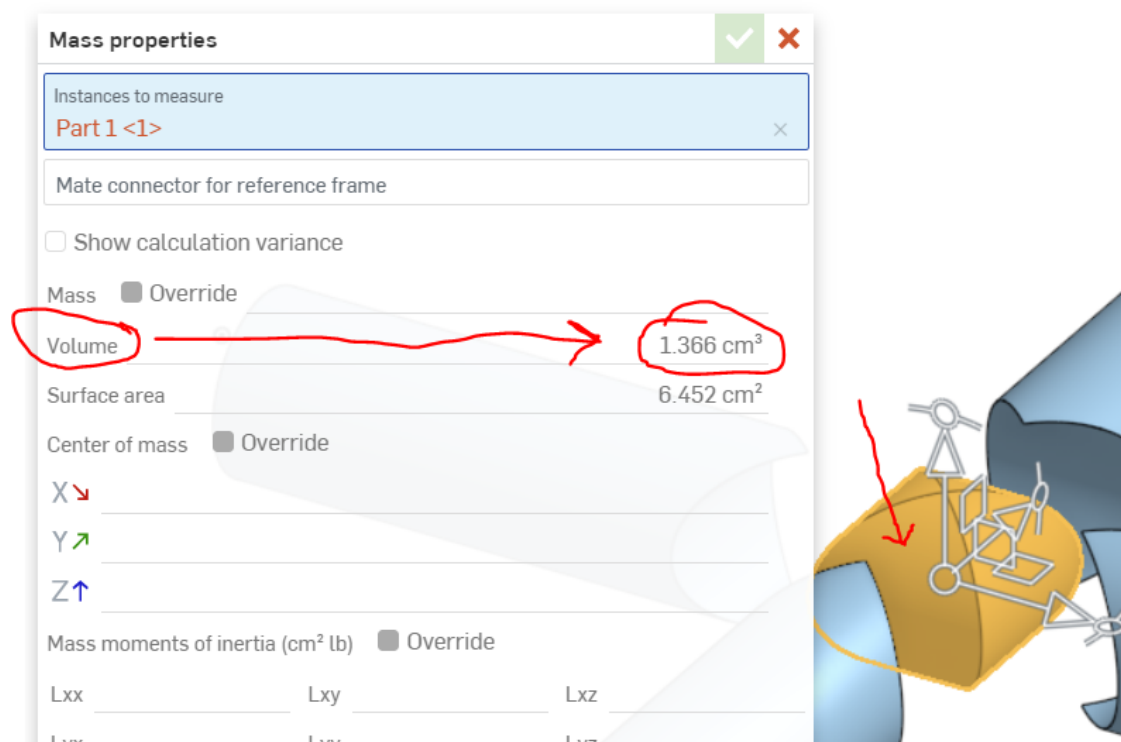
$$x = \sqrt{R^2 - z^2}$$

$$V_{TOT} = 8 \int_0^R x^2 dz = 8 \int_0^R (R^2 - z^2) dz$$

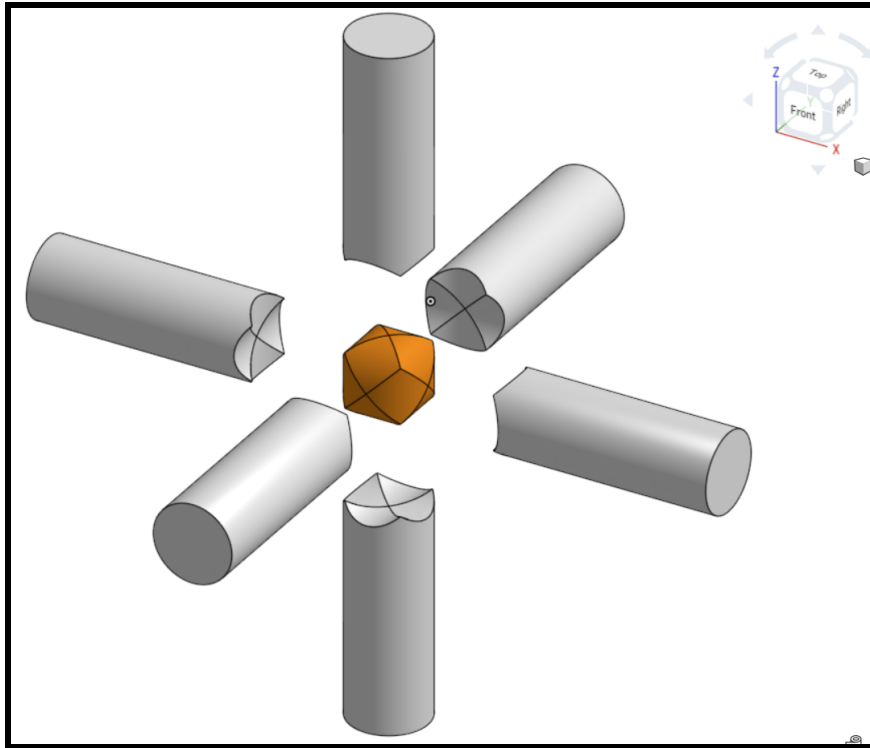
$$R = 0.635 \text{ cm}$$

$$V_{TOT} = 8 \int_0^{0.635} (0.635^2 - z^2) dz = 1.366 \text{ cm}^3$$

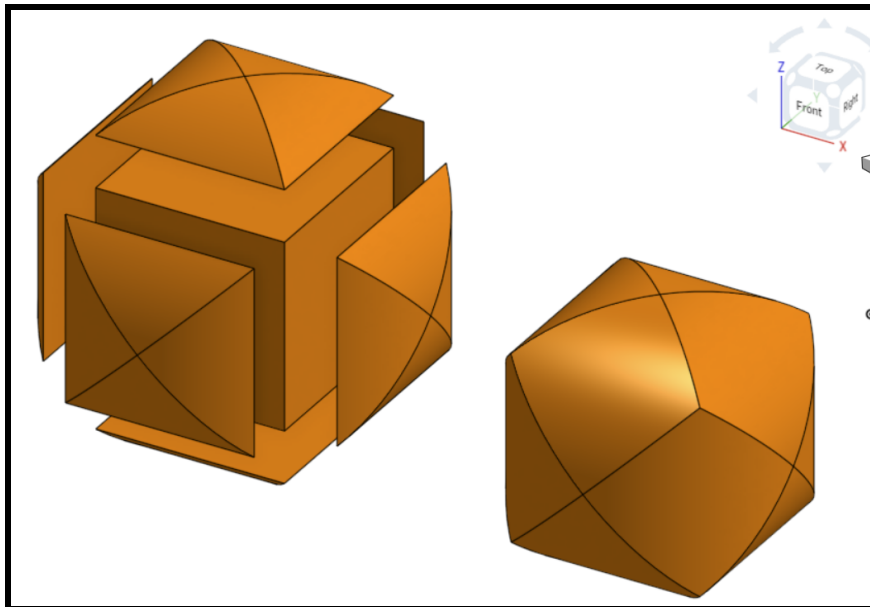
THIS ANSWER IS THE SAME AS THE THEORETICAL VOLUME CALCULATED BY THE ONSHAPE PROGRAM!!!



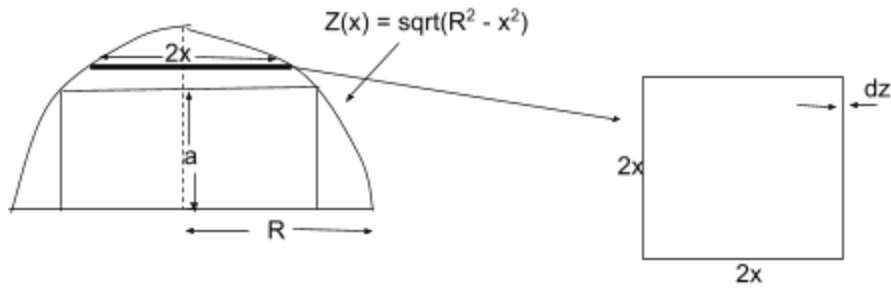
3D Intersect Slideaway Image...



3D Intersect Slice Up...



3D Intersect Analytical Volume Calculation...



$$z(x) = \sqrt{R^2 - x^2}$$

$$z^2 = R^2 - x^2$$

$$x = \sqrt{R^2 - z^2}$$

$$V_{TOT} = (2a)^3 + 6 \int_a^R (2x)^2 dz = (2a)^3 + 6 \int_a^R 4(R^2 - z^2) dz$$

$$R = 0.635 \text{ cm}$$

$$a = 0.449 \text{ cm}$$

$$V_{TOT} = (2(0.449))^3 + 6 \int_{0.449}^{0.635} 4(R^2 - z^2) dz = 1.999 \approx 1.2$$

THIS ANSWER IS THE SAME AS THE THEORETICAL VOLUME CALCULATED BY THE ONSHAPE PROGRAM!!!

